water & sanitation
Department:
Water and Sanitation
REPUBLIC OF SOUTH AFRICA

WATER RESEARCH COMMISSION

TERMS OF REFERENCE FOR A SOLICITED WRC PROJECT

THEME	**Water Availability**
TITLE	**Validation and verification of existing lawful water use: Development of a system for management of data, documents and workflows**
TOR NUMBER	**1010035**

Rationale

The Department of Water and Sanitation (DWS) is undertaking a validation and verification (V&V) process for existing water use. The primary purpose of V&V is to measure the extent of Existing Lawful Use (ELU) of water, as defined in sections 32-35 of the National Water Act. Validation is a technical process to confirm how much water was being used in the qualifying period preceding the 1998 National Water Act (NWA) coming into force. Verification is a legal process to determine the lawfulness of this water use.

This information will assist DWS in better managing the country's water resources and maximise the benefits associated with the outcomes of V&V. Among others, these benefits include updating the Water Use Authorisation and Registration Management System (WARMS) database, promoting a balance between supply and demand, providing certainty to ELU water users across the country, identifying unlawful water use, and providing a basis for compulsory licensing and water re-allocation.

Compulsory licensing, as provided for in sections 43-48 of the NWA, is intended to be used in areas which are, or are soon likely to be, stressed in terms of water quantity or quality, or where it is necessary to review prevailing water use to achieve equitable access to water. Once compulsory licencing is initiated in a particular area, all ELU entitlement holders and new potential water users in that area are required to apply for water use licences.

The Water Research Commission (WRC) is working with DWS to expedite the completion of V&V across the country and support the acceleration of compulsory licencing. The scope of work outlined in these terms of reference focuses on conceptualising, developing and deploying a system to manage and store the information generated by the V&V process. Part of this will involve automating elements of the V&V work to

improve workflow efficiency, data storage and integration with other relevant business applications in DWS and the Department of Rural Development and Land Reform (DRDLR). A particular focus of this integration work will entail examining options for process automation of compulsory licencing, by determining if further development is needed of the Electronic Water Use Licence Application and Authorisation System (eWULAAS).

The work of doing V&V property by property has been carried out in a decentralised and semi-standardised way by DWS regional offices, Catchment Management Agencies (CMAs) and service providers. The NWA spells out the technical and administrative processes that must be followed. These prescripts were supplemented by a document first produced by DWS in 2006 to provide further guidance on implementing V&V. There has been considerable variation between regions in how V&V has been implemented. The lack of prescribed national templates and standards and the absence of a single national repository for V&V data has resulted in data being generated and stored at regional level using a range of templates, formats and standards. Data is currently stored primarily in spreadsheets held by DWS regional offices and CMAs.

Combining the data in these inconsistent, and sometimes incompatible, formats into a single national repository will be challenging. For this reason, the task of compiling the existing data into a format suitable for import into the system is being undertaken by V&V experts separate from this project. The data will be made available to this project in MS Access format, using the same structure specified below for the system to be developed through this project (see System Requirements). This dataset can also be used as the testing data.

V&V has been completed for around 39 000 properties and is at various stages of completion for others. The system to be developed must make provision for approximately 500,000 records (i.e. individual properties/farm portions). Of these, V&V will ultimately be completed and data captured on the system for about 125 000 properties. The number of users who will register on the system, including water users, intermediaries, and DWS and CMA staff, is not expected to exceed 125 000.

Objectives

General

Conceptualise, develop, deploy and maintain a system for managing data, documents and workflows related to the validation and verification of existing lawful water use.

Specific

1. Examine the potential of process automation innovations to expedite V&V and compulsory licencing
2. Design, develop, deploy and stabilise a system incorporating these innovations
3. Migrate existing V&V data into the system
4. Provide post-implementation support and maintenance

5. Build capacity within DWS and CMAs for maintenance and use of the system

System requirements

1. Required functionality:
    a. Storage of data generated by the V&V process, according to the database structure outlined in Appendix B of the draft *Requirements for verifying the extent and lawfulness of existing water use*
    b. Storage of digitised copies of all documents generated through and relevant to the V&V process (e.g. section 35 letters, gazette notices, satellite imagery, Water Tribunal decisions, correspondence with water users).
    c. Process automation for each property where V&V has still to be completed, according to the business processes in Appendix C of the draft *Requirements for verifying the extent and lawfulness of existing water use*. This includes:
        i. automated generation of section 35 and other letters using standard templates provided by DWS
        ii. electronic approval of tasks, including digital signatures for section 35 letters
        iii. ability for water users and officials in DWS and CMAs to track process of section 35 processes
        iv. reminders to managers of due or overdue tasks
        v. validation rules to ensure that the workflow does not move to the next step until the mandatory information for the current step has been captured and checked
        vi. an audit trail of actions
2. The system must be deployed into and will be housed in the DWS IT environment from the development phase onwards to ensure the necessary integration with other DWS systems and compliance with minimum DWS system requirements.
    a. While the system can be developed independently, it must be built on the same platform as e-WULAAS. A key initiative of the DWS ICT Digital Strategy is developing applications on a common data platform, which standardizes the logical data model to enable related applications to operate on and share the same data. Additionally, the Common Application Platform aims to standardize and streamline the application landscape, allowing related business activities to function on a shared platform. e-WULAAS runs on C#, ASP.NET and SQL Server platforms.
3. The system must be accessible to water users and DWS and CMA officials via an online portal.
4. Integration with other relevant business applications:
    a. A two-way flow of data is required between the V&V system and the DWS Water Use Authorization and Registration Management System (WARMS).
    b. Integration with e-WULAAS should include provision for pre-populating water use licence applications in e-WULAAS from the V&V database in cases where the applications are for conversion of ELUs to water use licences.

   c. The feasibility of integration with relevant databases under the jurisdiction of DRDLR, specifically those maintained by the Surveyor General and/or Deeds Office, must be examined with the aim of enabling property information to be extracted and integrated into the V&V data.

5. User-based access roles
   a. Self-service functionality is required, to allow water users to register on the system via an online portal, check the details of water use that have already been captured on the system, update their contact details, download or electronically sign section 35 letters etc.
   b. Adequate safeguards are required to protect access to confidential water user data on the system and limit access to data depending on the category of system user.
   c. An approach is required that ensures uniformity and consistency in how water users who are already registered on other DWS business systems (e.g. WARMS, e-WULAAS) can access the V&V system.
   d. DWS/CMA officials also need to be registered and linked to user-based access roles.
   e. The requirements gathering phase at the start of the project will need to define the different groups of users and what rights they will have.
6. Reports generated by the system:
   a. The types of reports the system will need to generate for DWS head office, regional offices and CMAs will be determined during the requirements gathering stage of the project, along with the templates for these reports
   b. Ability to generate reports in these pre-defined templates must be built into the system
   c. Functionality must also be built into the system to allow users to perform customised searches and queries (e.g. through provision of user-friendly query builder tools)
   d. The system must be able to pull information from WARMS and e-WULAAS to generate reports (e.g. for which properties V&V is complete but not yet registered on WARMS)
7. Generation of user files:
   a. Generate electronic files per user as V&V progresses. The content of these files is specified in Appendix A of the draft *Requirements for verifying the extent and lawfulness of existing water use*. Provision must be made for files to be printed by the responsible authority as and when required

Deliverables

The following deliverables are indicative and may be tailored to suit the proposed approach:

1. Inception report including a comprehensive project implementation plan and gap analysis
2. Business requirements/functional specification
3. System design
4. Configuration and deployment of the application package in the DWS IT environment, including test plans, test cases, testing results and reports

5. Data migration tools, plans and progress reports
6. Training manuals, technical administration guides and user training sessions
7. Monthly support and maintenance reports
8. Completion of system handover to DWS
9. Final report including all project close-out documentation and recommendations for further development of eWULAAS to facilitate compulsory licencing

Framework and Methodology

The appointed contractor must adhere to applicable software development best practices as well as methodologies and frameworks aligned with the following DWS standards and guidelines:

1. Information Security Policy
2. Information Technology Product Standards
3. ICT Digital Strategy

Special conditions

In appointing a contractor to undertake this assignment, the WRC is required to make use of SITA Transversal Contract RFB1183/2022. Only service providers listed in RFB1183/2022 will be eligible for appointment. Further, service providers must be accredited in Gauteng for all the services listed in RFB1183/2022 that are relevant to these ToR. Compliance with these requirements must be indicated by completing the table in Appendix A and uploading it as part of the submitted proposal.

Furthermore, service providers must be Microsoft Gold Application Development certified partners.

Proposals from service providers that are not accredited through SITA Transversal Contract RFB1183/2022 for all the services listed in Appendix A, and that are not Microsoft Gold Application Development certified partners, will not be considered for evaluation.

For WRC to determine compliance of proposals with the above requirements, the following documents must be uploaded to the WRC Business Management System (BMS) along with the proposal:

1. Certification as a Microsoft Gold Application Development partner
2. Certificate or letter from SITA indicating accreditation under RFB1183/2022
3. Completed Annexure A of these ToR indicating accreditation in Gauteng for the specific services required for this assignment.

Notes:

1. An optional briefing session will be held as follows:

   Date and time: 24 June 2025, 10:00-11:30

   Venue: Online via Microsoft Teams: **Click here to join the meeting**

   Meeting ID: 328 968 771 944 5, Passcode: LK2Ye25P

2. Proposals must be submitted online via the WRC Business Management System (BMS). A user guide to BMS is available here. For technical queries regarding BMS, contact bms-support@wrc.org.za. **Closing date for submission of proposals is 16:00 on 7 July 2025**.

3. Prior to capturing a proposal on BMS, proposers should familiarise themselves with the guidelines for submission of research proposals. Particular attention should be given to the section in the guidelines on budgeting.

4. Project team composition must reflect the full range of experience and skills necessary to undertake this assignment. Proposals must include the details of all team members, their experience and expertise relevant to this assignment and their role in the project.

5. Proposers must have the necessary industry-recognised certification/s related to the technology proposed to be used for developing the system. Evidence of these certifications is to be uploaded as supplementary documentation when submitting the proposal.

6. Contact person for enquiries on these terms of reference: John Dini (johnd@wrc.org.za)

Time Frame:

Planned project start date: 1 October 2025

Live deployment of system and online user interface: 6-9 months from project start date

Provision of support and maintenance for the balance of the contract period

Project end date: 31 October 2026

Total Funds Available:

R4,500,000 including VAT

Appendix A

List of ICT services required for this ToR.

Proposers must be accredited for the following ICT services, listed according to the ICN numbers in the SITA RFB 1183/2022 transversal contract. The full description of these services is available at https://www.sita.co.za/content/services-2.

Indicate compliance by placing a tick (✓) in the applicable column for each listed ICN number. This table must be uploaded on BMS as part of the proposal submission.

Name of service provider: _____

ITEM NO	ICN NO	ICT SERVICES AND DEFINITIONS	ACCREDITED IN GAUTENG	NOT ACCREDITED IN GAUTENG
ICT Management Services				
1	81112011-0005	ICT Management - Project Management		
Business Solutions Delivery Services				
2	81112011-0018	Business Solutions Delivery - System Analysis and Design		
3	81112011-0019	Business Solutions Delivery - Business Solution Development		
Business Solution Implementation Services				
4	81112011-0038	Business Solution Implementation - Application / ICT/COTS Training		
5	81112011-0042	Functional Application Support/COTS/ICT Services		
Data Centre Services				
6	81112011-0058	Data Centre - Database Support and Maintenance		